

# CallRank

CallRank: Isolating What Changed in Earnings-Call Q&A for Sector Ranking and Cost-Adjusted Alpha

**Comparing each company's earnings-call Q&A with its own past calls ranks which sector ETFs will beat SPY. In a 24-month out-of-sample test, buying the top-ranked sector each month returned 42.1% a year after costs, against 24.0% for SPY.**

**.158**

**monthly sector-rank IC**

raw displacement; Newey–West  $p = .007$

**42.1% vs 24.0%**

**annual net return, Top 1 vs SPY**

unit direction; 5 bp one-way costs

**2.05 vs 1.17**

**Sharpe ratio, Top 1 vs SPY**

max drawdown 9.9% vs 18.8%

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# What changed in the call, not who is talking

A one-minute read for someone new to the topic. Slides 03–09 give the evidence.

## In plain terms

**The problem** Earnings-call Q&A holds timely information, but most of its language is what that company always says. A model trained on raw transcripts learns to recognise the company rather than what is new.

**The idea** Turn each call into a vector, subtract the same firm's own past-call average and keep the signed direction of the change. Average the changes by sector and rank the 11 US sector ETFs each month.

**The result** Over 24 out-of-sample months (Dec 2023–Apr 2026) the monthly sector-rank IC was .158; buying the top-ranked sector returned 42.1% a year after costs versus 24.0% for SPY, with about half the drawdown.

## MY ROLE · WHERE THIS COULD BE USED

**My role** Third author.

**Use case** A monthly sector-rotation signal from transcripts, and a general recipe for removing identity from text embeddings.

## Terms used in this deck

### EMBEDDING

A vector of numbers that represents a passage of text; here from Industry-BERT-SEC, a finance-tuned language model.

### SIGNED DISPLACEMENT

The vector difference between this call and the firm's own past calls; it keeps direction, unlike a distance.

### IC (INFORMATION COEFFICIENT)

Correlation between the predicted sector ranking and the realised one; 0 is a coin flip.

### SECTOR ETF

A fund tracking one US sector, such as Technology; SPY tracks the whole S&P 500.

### OUT OF SAMPLE

Decisions made on months the model never saw in training; the encoder was released before the test began.

### NEWBY-WEST p-VALUE

A significance test that allows for overlapping, autocorrelated monthly returns.

# Isolate what changed, not who is talking

Earnings-call Q&A is timely, but most of its language repeats what the same company always says.

## Why it matters

- Sector indicators confirm shifts after they begin. The less-scripted Q&A adds information beyond the announcement.
- Calls from the same company cluster in embedding space, so a model can learn the issuer instead of what is new.
- Distance throws away direction: stronger demand and rising cost pressure can sit at the same distance from past language.
- Pooling firms into sectors lets large issuers and long transcripts dominate.

## WHAT THIS MEANS OUTSIDE THE PAPER

**Each firm vs its own past** what is left is what changed, not who is speaking.

**Frozen before the test** heads fit through 2020, then unchanged for 24 decisions.

**Generalizes to** any embedding where identity dominates the signal you want.

## Data & setup

### TRANSCRIPTS

Financial Modeling Prep; 22,897 parsed → 21,421 locked calls

### MARKETS

11 U.S. sector ETFs vs SPY; Norgate constituents; Yahoo Finance prices

### OUT OF SAMPLE

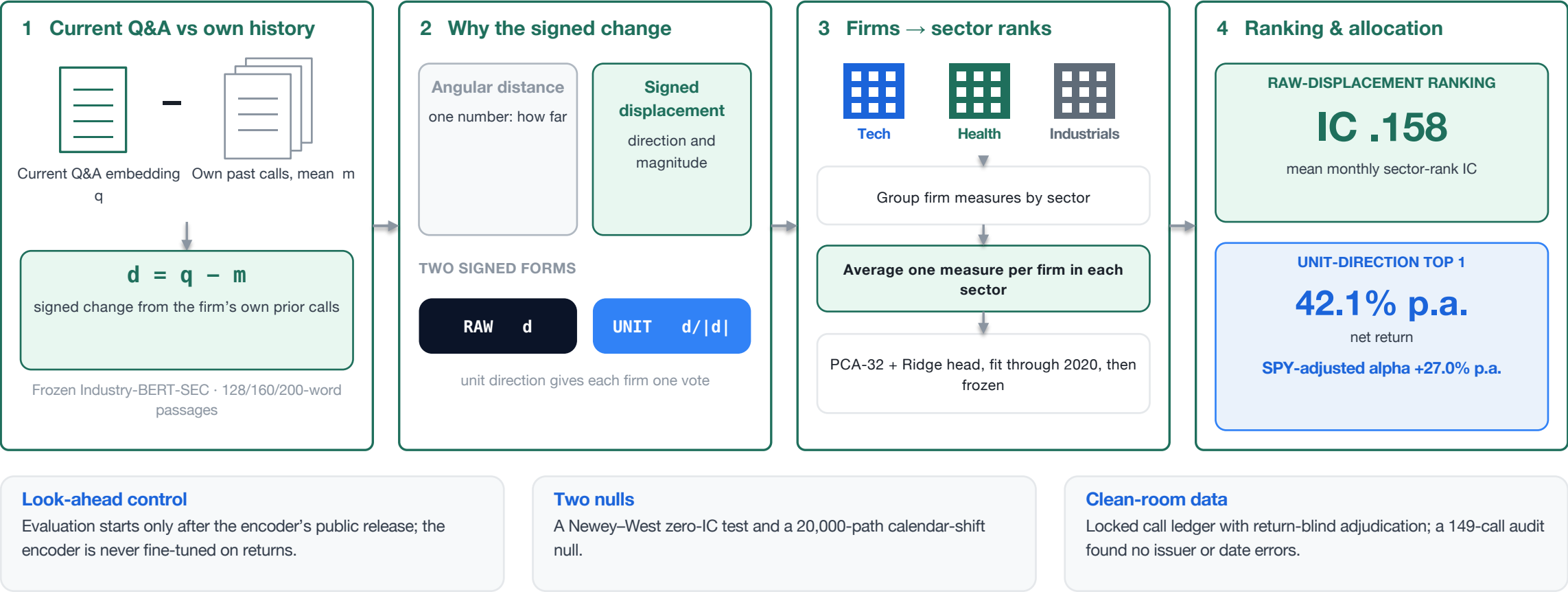
24 monthly decisions, Dec 2023–Apr 2026, after the encoder's release

### TRAINING

Ranking heads fit only on outcomes ending by Dec 1, 2020, then frozen

# From a firm's Q&A change to a sector bet

Each call is compared with the same firm's past calls; the signed changes are averaged by sector and ranked.

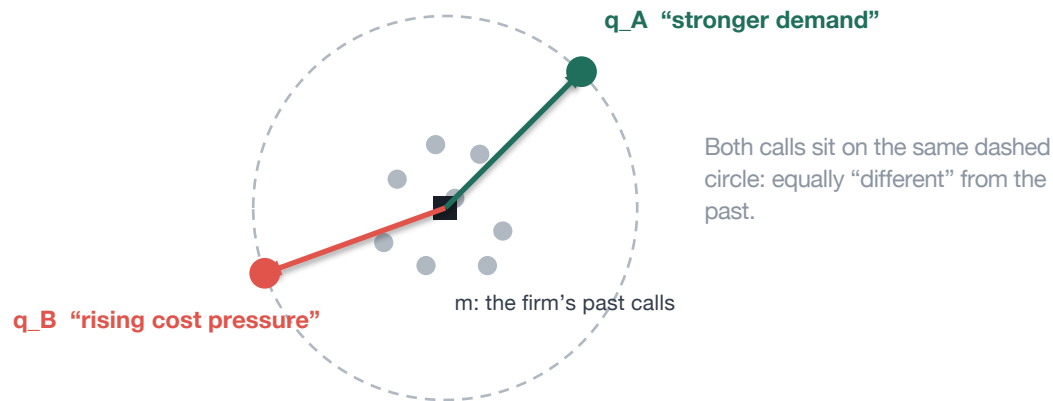


Redrawn and simplified from the paper's Figure 1.

# Why the signed change, and why each firm's own history

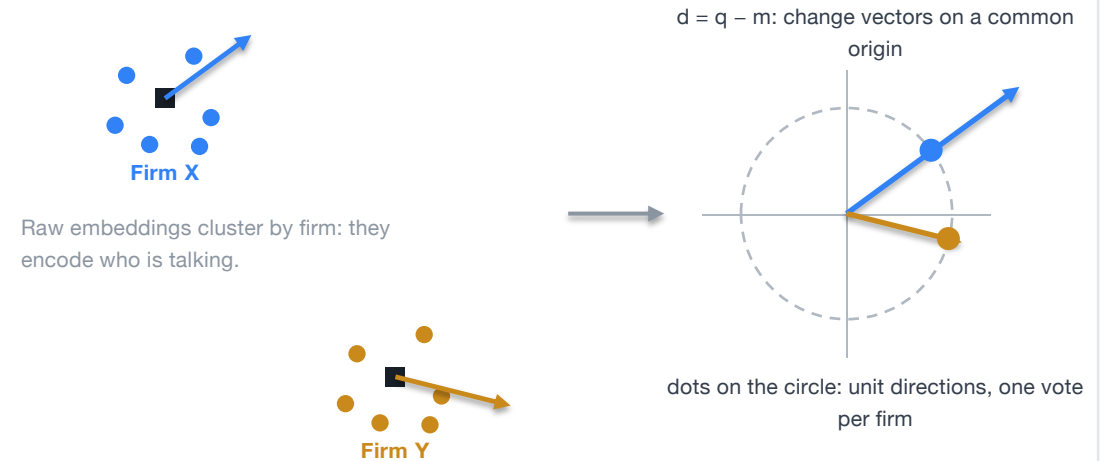
Distance alone cannot tell opposite news apart, and raw embeddings mostly encode who is speaking.

## Distance hides direction



A distance feature scores both calls the same. Signed displacement keeps the direction, so the two calls send opposite signals.

## Subtract each firm's own history



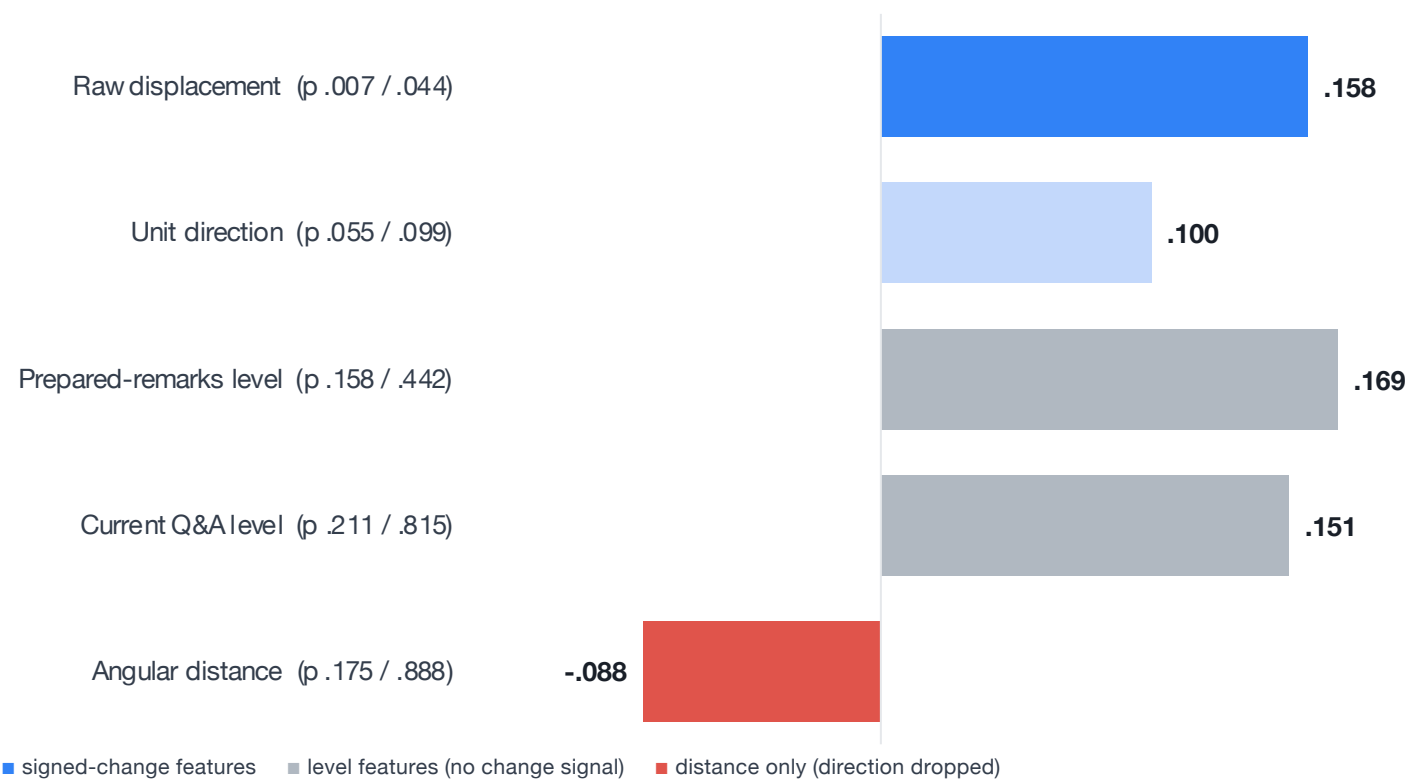
Subtracting the firm's own mean removes the part of the embedding that just identifies the firm; what remains is what changed.

Illustration of the idea in the paper's §3 and Eqs. 1–5; the points are schematic, not data.

# The signed change carries the signal, not the company name

Two level features score as high as the winner and both fail the tests. Only raw displacement survives both. p-values are Newey–West / calendar-shift.

## Sector-rank IC



Source: CallRank Tables 2–3 and Fig. 2.

### What it shows

- IC is the correlation between the predicted ranking and what actually happened. 0 is a coin flip; .158 monthly is a strong sector signal.
- Raw displacement is the only representation that passes both nulls. Level features with a similar IC fail both.
- Angular distance, which drops direction, scores  $-.088$ .
- Conditioning on each firm’s own history cuts issuer retrieval from 41.4% to 3.2%: the vector carries the change, not the name.
- Separately, masking company, ticker and speaker names from the text still leaves IC .191, while a body-swap placebo falls to  $-.057$ .

# After costs, the top-ranked sector beats SPY

Top 1 allocation (unit direction), net of 5 bp one-way costs; decisions Dec 2023–Apr 2026.

| Portfolio             | Ann. net return | Sharpe | Max drawdown | Beta |
|-----------------------|-----------------|--------|--------------|------|
| CallRank Top 1        | 42.1%           | 2.05   | 9.9%         | 0.55 |
| SPY                   | 24.0%           | 1.17   | 18.8%        | 1.00 |
| LM Q&A tone residual  | 20.1%           | 0.95   | 17.6%        | 0.62 |
| 1/N sectors           | 17.2%           | 0.96   | 15.3%        | 0.73 |
| CallRank Bottom 1     | 13.0%           | 0.57   | 24.4%        | 0.77 |
| 42-day momentum       | 11.6%           | 0.47   | 20.3%        | 0.66 |
| 63-day low volatility | 7.3%            | 0.27   | 11.1%        | 0.34 |

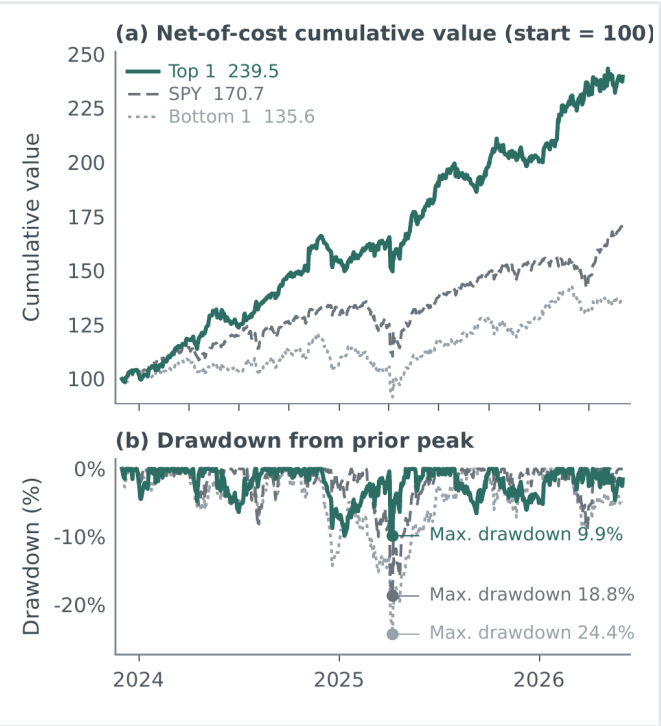
## What it shows

- Alpha: 27.0% SPY-adjusted, 24.0% CAPM, 20.3% Fama–French 5 + momentum.
- At 20 bp one-way costs: still 39.6% net and 24.8% SPY-adjusted alpha.

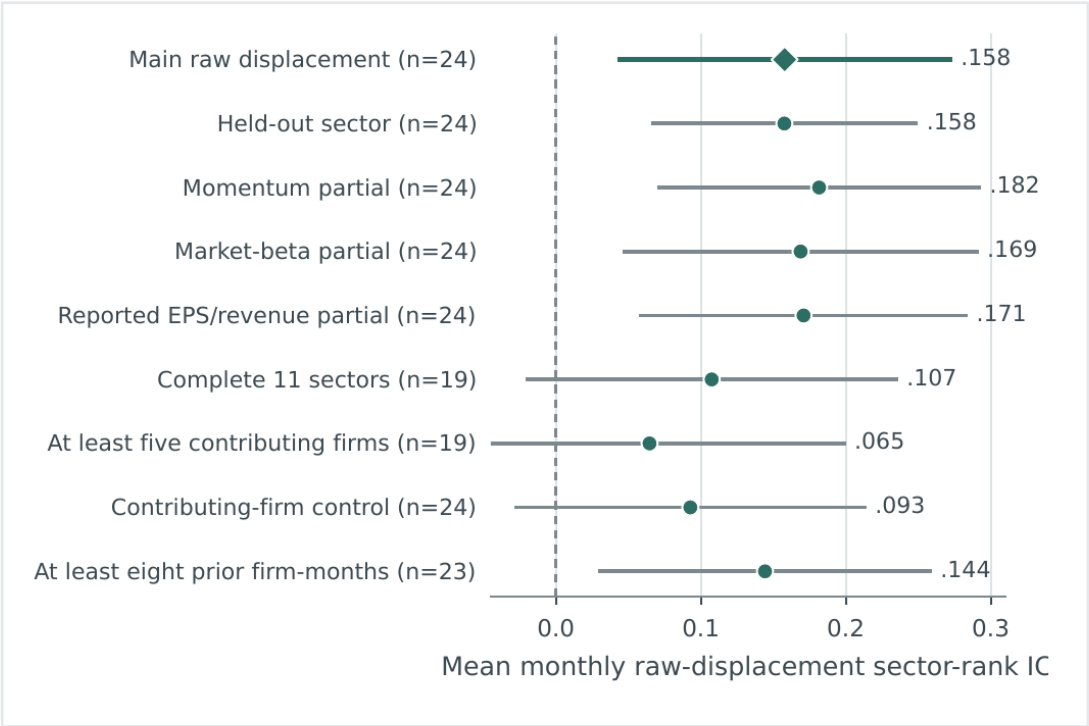
Source: CallRank Table 4, Fig. 3 and §5.4.

# The return path and robustness, as published

Original figures from the submitted paper, shown as published. Numbers on earlier slides are read from its tables and text.



**Paper Figure 3:** Net-of-cost value and drawdown, Top 1  
Growth of 100: 239.5 for Top 1 vs 170.7 for SPY; drawdown 9.9% vs 18.8%.



**Paper Figure 4:** Robustness of raw-displacement sector-rank IC  
All nine point estimates are positive; six of nine 95% Newey–West intervals exclude zero.

# What held, and what the paper does not claim

The two headline numbers come from two different representations; the paper says so explicitly.

## Checks that held

- Against 2,000 random sector paths with the same accounting, CallRank sits in the favorable tail for return, Sharpe, alpha and drawdown (all  $p \leq .009$ ).
- Momentum, market-beta and reported-earnings partial ICs stay at .169–.182.
- Picks span 10 sectors and change in 16 of 23 transitions; Top 1 beats Bottom 1 in every calendar year.
- Post hoc: replacing Utilities when it ranks first still gives 29.3% net.
- Industry-BERT-SEC is the only matched encoder that rejects both nulls at a common passage length.

## What the paper does not claim

- The IC headline (.158) uses raw displacement. The portfolio uses unit direction, whose own IC is .100 (NW  $p .055$ ). The paper claims neither dominates.
- It predicts which sector leads, not why, and not individual stocks.
- Only 24 monthly decisions; borrow costs excluded.
- The encoder's pretraining cutoff is undisclosed; starting evaluation after its release is the control.

Code (anonymous review repository): [anonymous.4open.science/r/callrank-icaif26-v11-review-artifact-740C](https://anonymous.4open.science/r/callrank-icaif26-v11-review-artifact-740C)